

Chapter 12 Summary — Sequence Models and LLM Agents in Strategic Settings

Research on the possibilities for applying Artificial Intelligence in computer games

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12 Chapter 12 — Sequence Models and LLM Agents in Strategic Settings

12.1 Why reframe reinforcement learning as sequence prediction

Classical RL optimises: estimate a value function, back up rewards, improve the policy. The Decision Transformer (Chen et al., 2021) proposes something different — treat the whole problem as **supervised sequence modelling**. Feed a GPT-style model a trajectory of triples

$$(\hat{R}_1, s_1, a_1, \hat{R}_2, s_2, a_2, \dots)$$

where $\hat{R}_t = \sum_{t' \geq t} r_{t'}$ is the *return-to-go*, and train it to predict a_t from everything up to s_t . There is no Bellman backup and no policy-improvement step. At inference you simply **condition**: ask for a high return-to-go and the model produces the actions that historically preceded such returns.

The appeal for this dissertation is that it works entirely **offline**, on a fixed dataset — the regime of recorded game logs, where self-play is unavailable.¹

¹Chen et al., *Decision Transformer: Reinforcement Learning via Sequence Modeling*, NeurIPS 2021 (arXiv:2106.01345).

12.2 The luck-versus-skill trap

The reframing hides an assumption: that the return-to-go is something the agent *earned*. In a stochastic environment it is not. Paster et al. (2022) show that return-conditioned policies systematically chase luck, and the smallest possible example makes it obvious.

Consider a one-step bandit. Action A pays 0.5 deterministically. Action B pays 1.0 with probability 0.4 and 0 otherwise, so $\mathbb{E}[B] = 0.4 < 0.5$. Action A is EV-optimal. But the *only* way a trajectory ever achieves a return of 1.0 is to take B **and get lucky**. Condition on “return = 1.0” and you recover exclusively B-trajectories:

$$P(a = B \mid \hat{R} = 1.0) = 1.00 \quad (\text{measured in own simulation})$$

The return-conditioned learner confidently selects the action with the *worse* expected value. This is not a bug in the model; it is what conditioning on an outcome means when the outcome is partly noise.²

12.3 ARDT — conditioning on what you can guarantee

The fix is to condition not on the return that *happened* but on the return the protagonist can **guarantee against a worst-case opponent** — the minimax return-to-go. ARDT (Tang et al., 2024) estimates it with **expectile regression**, which minimises the asymmetric squared loss

$$L_{\text{ER}}^{\alpha}(u) = \mathbb{E}_u[|\alpha - \mathbf{1}(u > 0)| \cdot u^2]$$

whose limits give the operators you need: $\alpha \rightarrow 0$ recovers the **minimum** and $\alpha \rightarrow 1$ the **maximum**. The trajectories are relabelled with the estimated minimax return and a standard DT is trained on the relabelled data.

Two details of the published method matter, and both were confirmed by reading the source rather than the summary. First, the pessimistic side is **low** α (the chapter’s original plan wrongly called $\tau = 0.9$ pessimistic); the paper itself runs $\alpha = 0.01$. Second — the important one — the relabel target is

$$\tilde{R}_t = \tilde{Q}_{\nu}(s_t, a_t)$$

a state-**action** value produced by two *coupled* estimators alternately fitted with α and $1 - \alpha$ losses. A state-only value $V(s)$ cannot distinguish “this state is bad” from “*this action* in this state is bad”, which is exactly the discrimination the method relies on.

²Paster, McIlraith & Ba, *You Can’t Count on Luck: Why Decision Transformers and RvS Fail in Stochastic Environments*, NeurIPS 2022 (arXiv:2205.15967).

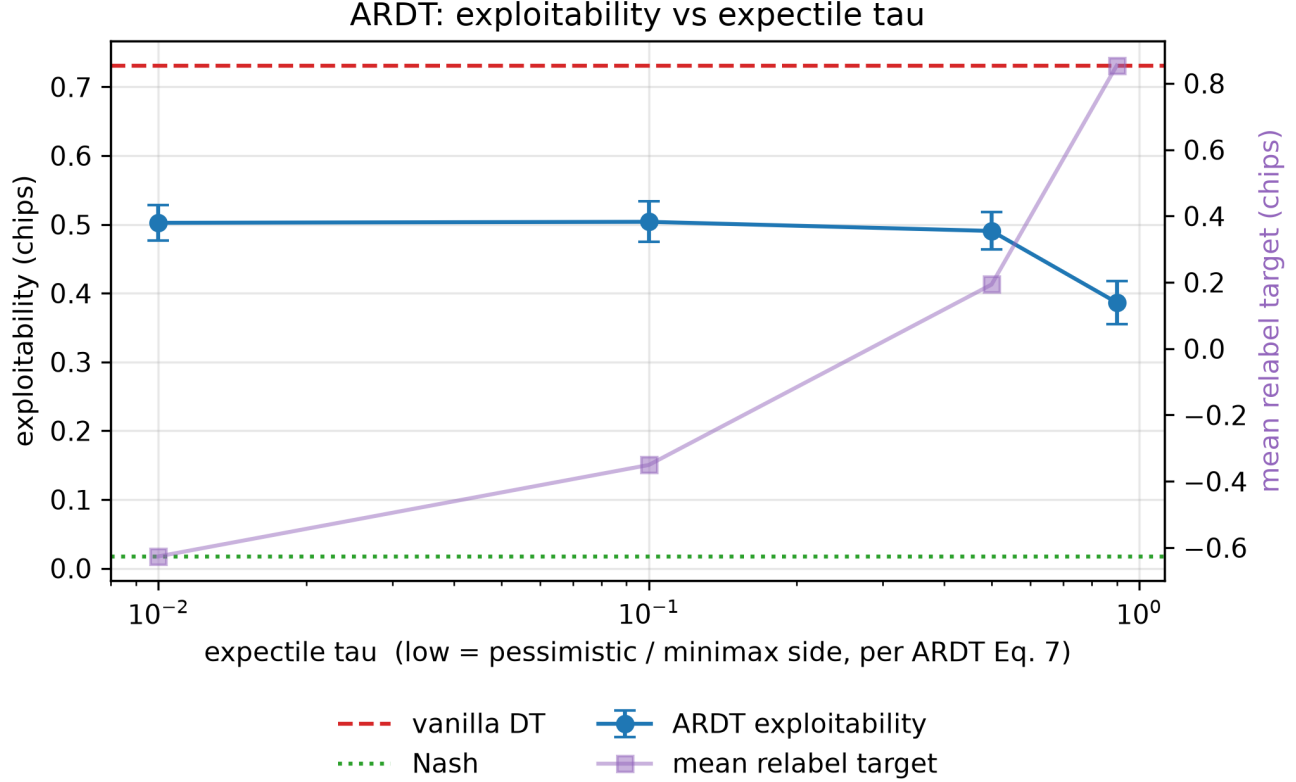


Figure 1: ARDT exploitability against the expectile parameter τ ; on the second axis, the mean relabel target. Means over three seeds (for comparison: vanilla DT 0.731, equilibrium 0.017 chips).

Measured on Kuhn with a deliberately simplified state-only proxy, the relabel target moves monotonically with τ as theory requires — but exploitability is *lowest* on the optimistic side. The most likely cause is the missing action argument; a $\tilde{Q}(s, a)$ variant has not been run, so this neither refutes nor confirms ARDT.³

12.4 What return conditioning actually does in poker

Kuhn Poker’s equilibrium and exploitability are exactly computable (see Chapter 2),⁴ so the question can be settled rather than argued. The result is that conditioning **changes** the policy substantially but does not **steer** it: exploitability is flat across the range of real target returns, except for a sharp spike (to ≈ 1.9 chips) at one specific value — the modal payoff, which in Kuhn is the payoff of folding.

On the same data plain behavioural cloning reaches 0.055 chips of exploitability — close to Nash (0.016) — against 0.799 for the DT: return conditioning actively destroys information (the DT value varies between

³Tang, X., Marques, A., Kamalaruban, P. & Bogunovic, I., *Adversarially Robust Decision Transformer*, NeurIPS 2024 (arXiv:2407.18414) — Section 3 and Algorithm 1.

⁴Kuhn, H. W. (1950). “A Simplified Two-Person Poker.” In H. W. Kuhn & A. W. Tucker (Eds.), *Contributions to the Theory of Games*, Vol. I (Annals of Mathematics Studies 24), pp. 97–103. Princeton University Press. DOI 10.1515/9781400881727-010.

retrains, 0.67–0.80).

The natural explanation is that in a four-payoff game the *magnitude* of the return encodes which betting line was played, while its *sign* encodes who held the better card. Conditioning therefore selects the shape of the hand rather than the quality of the play.

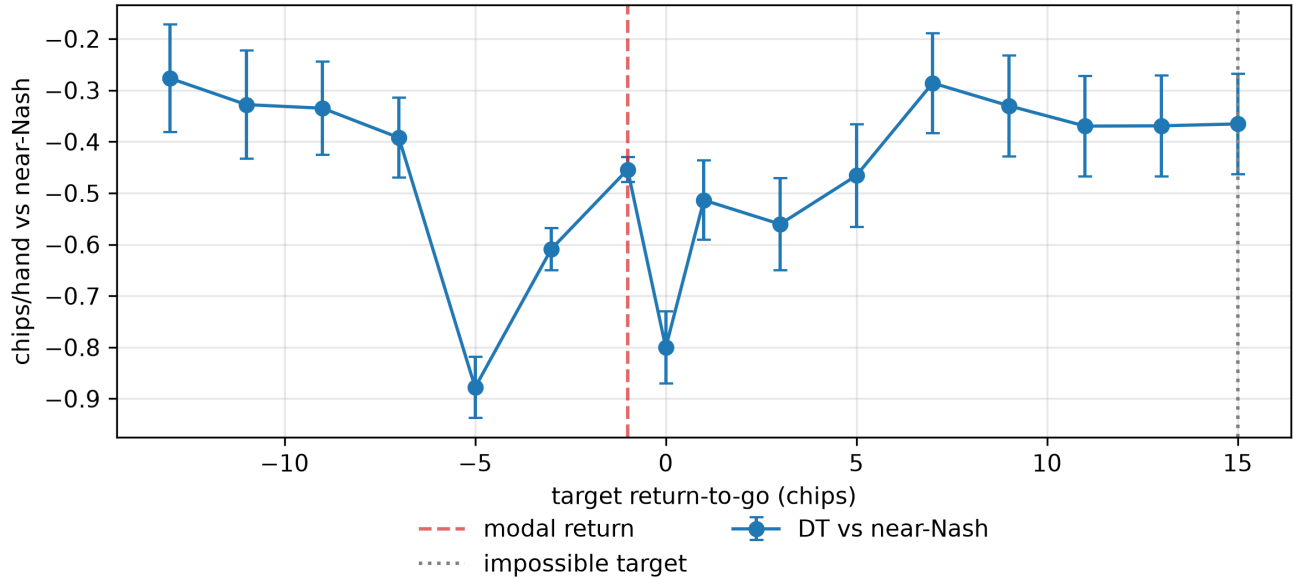


Figure 2: Decision Transformer performance on Leduc Hold'em (chips per hand against a near-Nash opponent) against the conditioned target return, with standard errors (Pearson $r = +0.062$). Dashed line: the most common return (-1); dotted line: the impossible target ($+15$).

Leduc Hold'em tests that explanation with fifteen payoff values instead of four, two betting streets and a board card. **The notch at the modal return disappears as the explanation predicts — but steering still does not appear** (Pearson $r = +0.062$); two unexplained dips remain, at 0 (the second most common return) and at -5 . So the payoff-alphabet account explains the collapse but not the failure. The failure is more fundamental: in a zero-sum imperfect-information game the realised return is dominated by the opponent's private card and actions, which the protagonist does not control. No amount of payoff granularity makes an uncontrollable signal steerable.

This is the chapter's main conclusion: on recorded logs, a return-conditioned Decision Transformer is the wrong instrument, and the value of ARDT's relabeling is that it replaces a conditioning target the agent does not control with one that depends mainly on its own actions.

12.5 Language models as strategic agents

The second paradigm skips training altogether: describe the rules in English and let a language model play. Three open-weight models were run locally (LM Studio 0.4.20 on an RTX 5090, temperature 0.7; the quantisation of the weights was not recorded): gpt-oss-20b, Qwen2.5-7B-Instruct and OpenThinker3-

7B (a reasoning fine-tune of Qwen2.5-7B-Instruct). Unless stated otherwise, the results below are for Qwen2.5-7B-Instruct; all numbers are the author’s own measurements. Similar experiments on Kuhn, Leduc and Limit Hold’em show that larger models also fail to approach equilibrium play and that their actions often diverge from their stated reasoning⁵; prompting explicit reasoning about the opponent lets GPT-4 adapt its play.⁶ Measured against the same exact ruler, LLMs are **honestly exploitable** — but the interesting result is *where* they fail.

They get hand **ranking** right and mixing **frequencies** wrong. Every real model tested value-bets the King with probability 1.00 where the CFR equilibrium mixes at 0.68, and none bluffs the Jack under a plain prompt in 24 samples per information set. (Kuhn’s first-player equilibria form a one-parameter family — bluff the Jack with $\alpha \in [0, 1/3]$, bet the King with $3\alpha^4$ — so each behaviour alone is an equilibrium action; only the combination is not.) A 7B model (Qwen2.5-7B) matches gpt-oss-20b — a mixture-of-experts model with 21B total but only 3.6B active parameters per token — so three models on one toy game say nothing general about scale.

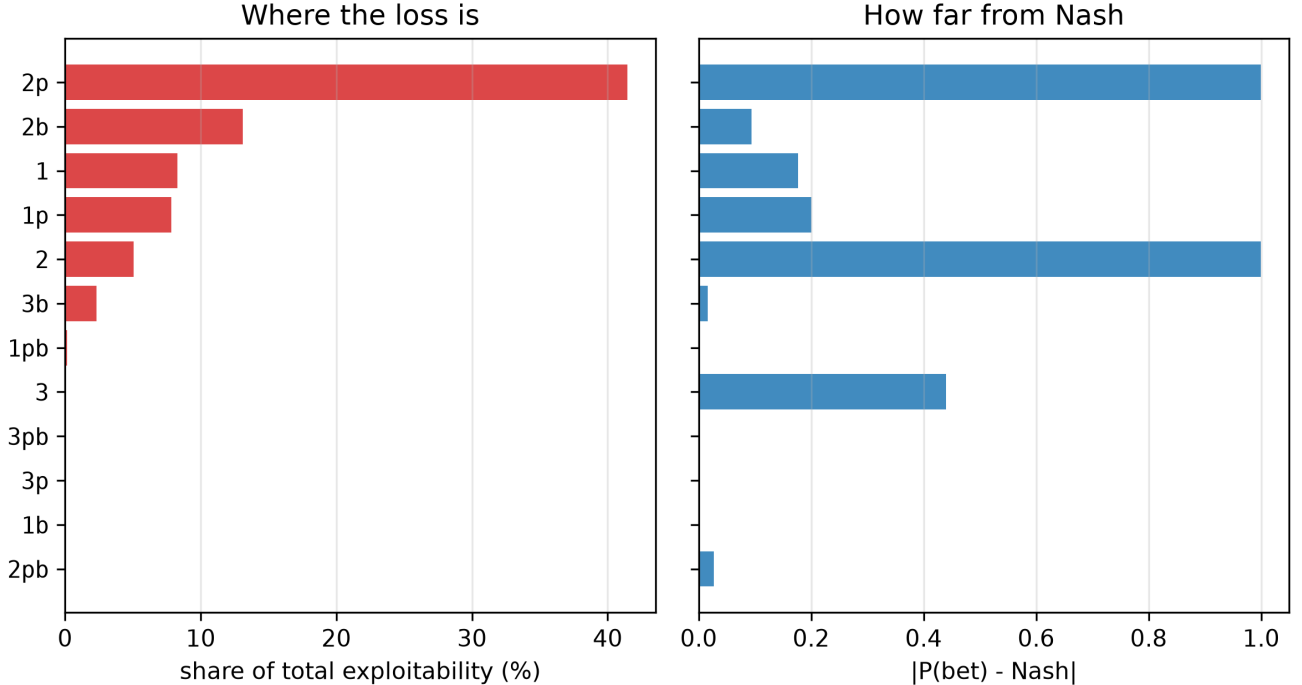


Figure 3: Each information set’s share of the total exploitability of Qwen2.5-7B-Instruct in Kuhn Poker (0.357 chips), beside its deviation from Nash equilibrium ($1/2/3 = \text{Jack/Queen/King}$; p = pass, b = bet in the history). The two orderings barely correspond.

Decomposing the loss per decision overturns the obvious reading. The King “failure” — the largest visible deviation from equilibrium — costs **0.1%** of the total loss, because over-betting a hand that is

⁵Lin, M., Dai, E., Liu, H. et al., *How Far Are LLMs from Professional Poker Players? Revisiting Game-Theoretic Reasoning with Agentic Tool Use*, ICLR 2026 (arXiv:2602.00528).

⁶Guo, J., Yang, B., Yoo, P., Lin, B. Y., Iwasawa, Y. & Matsuo, Y., *Suspicion-Agent: Playing Imperfect Information Games with Theory of Mind Aware GPT-4*, 2023 (arXiv:2309.17277).

never behind is nearly free. A single Queen decision costs **41.4%**. Deviation magnitude and cost are almost uncorrelated, which is the concrete argument for reporting per-decision diagnostics rather than a single exploitability number.

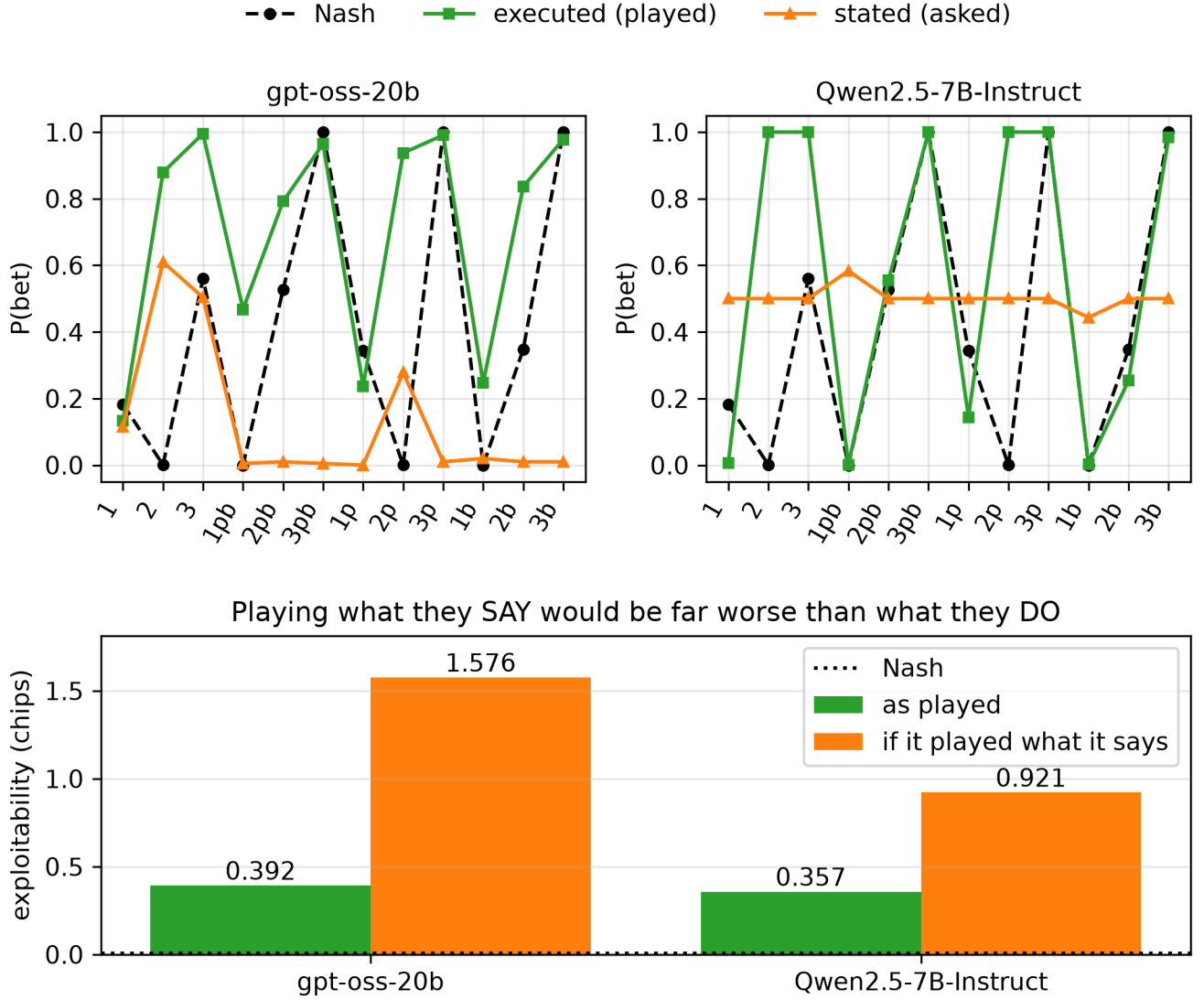


Figure 4: Stated (when asked) and actually played betting frequencies of gpt-oss-20b and Qwen2.5-7B-Instruct at each information set, with equilibrium for reference (top; mean absolute deviation from equilibrium, stated vs played: 0.43 vs 0.33 and 0.35 vs 0.25), and the exploitability of the played strategy and of a strategy built from the stated frequencies (bottom).

Asking a model what it intends is *worse* than measuring what it does. Scored as strategies, the frequencies these models **state** are 2.6× and 4.0× more exploitable than the strategies they actually **play**. One model answers “50%” almost everywhere; the other answers “near 0%” almost everywhere, including with the strongest hand. Behavioural probing is not merely more convenient than introspection here — it is more accurate (own measurement).

12.6 Exploitation, adaptation, and the limits of the toy game

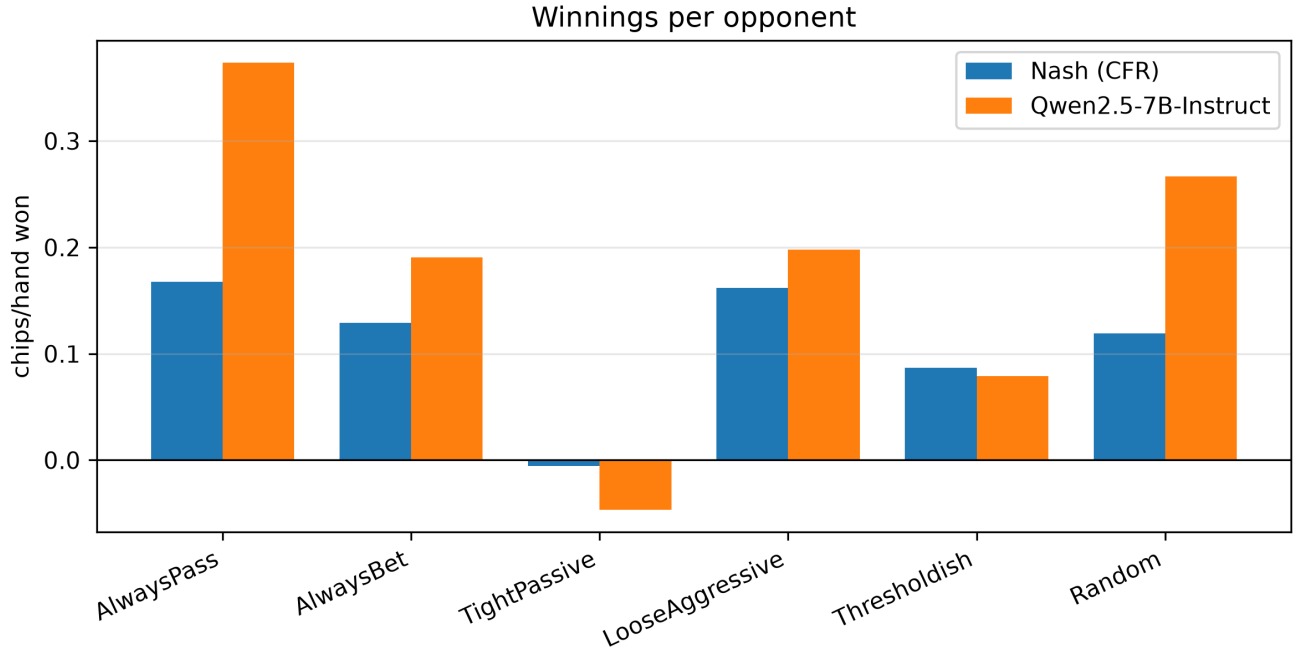


Figure 5: Winnings against each opponent in the zoo: Qwen2.5-7B-Instruct wins 0.177 chips per hand on average (equilibrium: 0.110) at an exploitability of 0.357 chips (equilibrium: 0.006).

Exploitability measures how a perfect adversary punishes you; it says nothing about how well you punish weakness. Measured against a zoo of deliberately exploitable archetypes, the language model (Qwen2.5-7B-Instruct) **wins 0.177 chips per hand against 0.110 for equilibrium play (61% more), while its exploitability is 0.357 chips against 0.006** — and nearly 90% of the advantage comes from the passive and random opponents. Against the two most competent archetypes it does *worse* than equilibrium. That is the safe-exploitation trade-off of the thesis’s second contribution, observed rather than assumed.

Given the last 20 hands of the session in context, the model (Qwen2.5-7B-Instruct) does **not** adapt: against a trivially passive opponent it captures 83% of the available exploitation in the first half and no more in the second (0.824 vs 0.827); the other two opponents are underpowered. What looks like opponent modelling is a fixed loose-aggressive prior. Told the opponent’s type in the prompt, however, the models shift their bluffing by up to 0.92 — what is missing is inference from history, not responsiveness to a description.

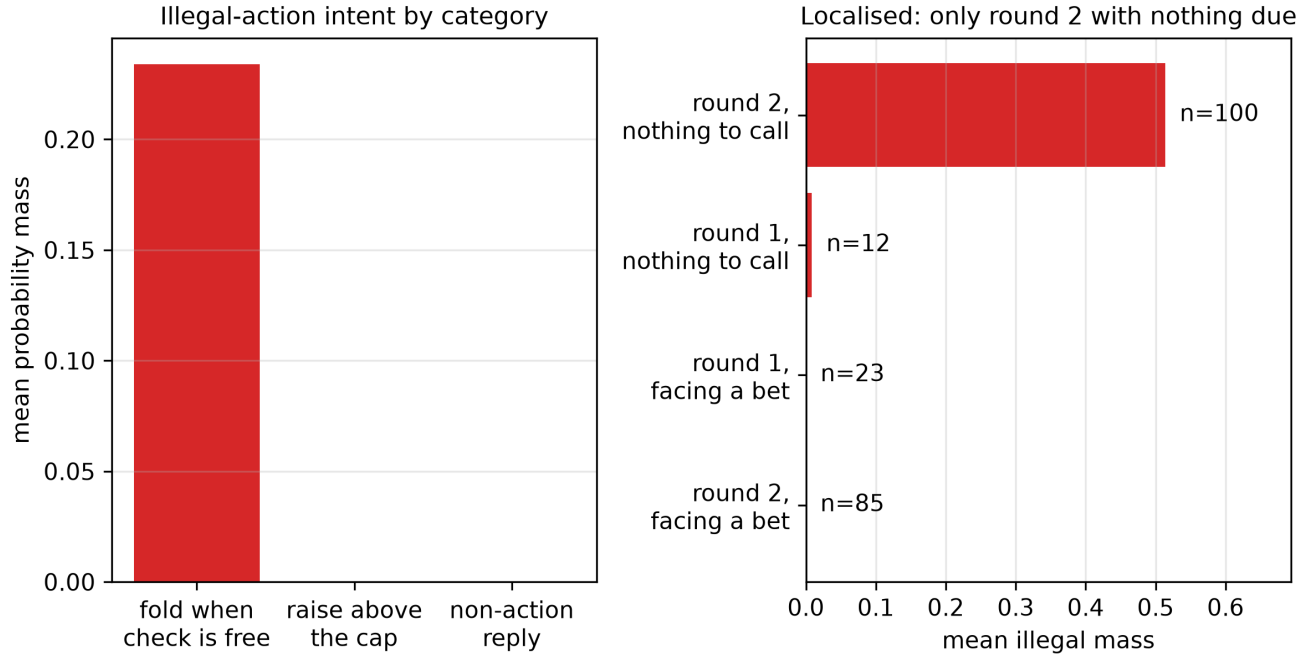


Figure 6: Illegal-action intent on Leduc Hold'em (Qwen2.5-7B-Instruct) by category and by situation (on average 0.234 of the probability mass over 220 information sets): almost all of it is folding when a check is free, in the second betting round.

Finally, the toy game flatters these models. One street and one board card later, the LLM is statistically indistinguishable from the Decision Transformer, loses on average to weak opponents (winning only against the two passive types), and wants to take an illegal action with roughly a quarter of its probability mass. That failure is a *single* misconception rather than diffuse confusion — it wants to **fold when checking is free**, almost exclusively on weak unpaired hands against a high board, while never once attempting to raise past the betting cap. Any optimism drawn from the toy game should be read as a property of the toy game.

12.7 Key takeaways for the thesis synthesis

- **Conditioning on a quantity the agent only partly controls does not steer it.** Observed on two toy games and consistent with Paster et al.'s Theorem 2.1; it is why ARDT's relabeling, not the Decision Transformer itself, is the more promising route for recorded poker logs — provided chance events (the deal) are handled, where ARDT's minimax target is no longer exact.⁷
- **The published ARDT relabels with a state-action value.** Reproducing the method's benefit most likely requires $\tilde{Q}(s, a)$, not $V(s)$ — the next step, still to be tested.
- **A scalar score ranks methods but does not diagnose them.** One decision carried 41.4% of the loss while the most conspicuous deviation carried 0.1%.

⁷Tang, X., Cheng, Z. & Kumar, S., *Robust Adversarial Reinforcement Learning in Stochastic Games via Sequence Modeling*, Reliable ML Workshop @ NeurIPS 2025 (arXiv:2510.11877) — proposes CART; notes that ARDT's minimax target is exact only for deterministic transitions.

- **Measure behaviour, not self-report.** Verbalised strategy was substantially worse than played strategy for every model tested.
- **The model did not infer its opponent from in-context history** (one 7B model, a 20-hand window) but did respond when told the opponent’s type, so the opponent model must be built explicitly and its output passed to the policy — directly relevant to the first thesis contribution.
- **Measurement protocol is part of the result.** At greedy decoding a language model plays a pure strategy and every measured frequency degenerates to 0 or 1; four separate findings in this chapter turned out to be measurement artefacts, each caught by a cheap check against an exactly computable quantity.